percent of its income in rent,[32] against 31.2 in HUD's Picture of Subsidized Households, and 39.2 percent of voucher holders live mainly on wages, against 28.0; the others are mostly retired or unemployed. Housing takes a large share of poor households' income, so the consumption Treatment First puts at stake when it conditions the voucher is of a similar size in the model and in the data.

Block 4: Income and wealth distributions. The lowest-income fifth of households receives 3.40 percent of income in the model, against 2.71 in the SCF, and holds 0.27 percent of net worth, against −0.23; the bottom half holds 6.3 percent, against 2.2. Both policies act on the households that receive vouchers, the poorest, and the model captures their relative poverty broadly in line with the data.

Block 5: Drug price elasticities. The calibration targets neither the participation nor the intensive elasticity of drug demand. The model's participation elasticity is −1.32, somewhat larger in absolute value than the estimates of about −0.3 to −0.9 in Saffer and Chaloupka (1999) and Grossman and Chaloupka (1998), and its intensive elasticity, over all users, is −0.63, in the middle of the estimates of about −0.3 to −0.8 in Grossman and Chaloupka (1998) and Olmstead et al. (2015). These elasticities are the response of drug use to its price, which is separate from the response to Treatment First itself. The latter, with the drug price held fixed, is matched to the evidence on impact when the detection rate is calibrated (Section 6). Given that, a larger price response can only work against Treatment First: as Treatment First lowers the drug price, more households return to using, and the more elastic participation is, the more of them do. A welfare gain from Treatment First is therefore conservative in this respect.

Block 6: Transmission. A parent's SUD and homelessness also carry over into its child's adult life, in the model as in the data. A child of a parent with a SUD has 1.46 times the odds of acquiring one in the model, within the range of 1.44 to 2.51 in Mellentin et al. (2016) and Yoon et al. (2013). An adult who is homeless has 6.23 times the odds of having had a homeless parent, above the 1.8 to 2.4 of Bassuk et al. (1997).[33]

Transmission in the model is therefore at the low end of the evidence on the SUD, the margin Treatment First acts on, and above it on homelessness, the margin Housing First acts on. With the penalty switched off, in the economy without the externalities of Section 6.2, the SUD odds ratio is 1.02, so the transmission runs through δ_d. The value of δ_d that would put the model in the middle of the range is above 0.24, so the calibrated value is likely at the low end of what the evidence supports. The model loads the externality on children towards Housing First more than Treatment First, so if Treatment First still raises welfare in the experiments below, that result is conservative.

Furthermore, in the model, 5.82 percent of children live with a parent who has a SUD; in NSDUH 2024, 4.6 percent of children whose parents are aged 26 or older live with a parent who has a moderate or severe drug use disorder. The externalities on the child in the model therefore reach a share of children similar to that in the data.

Block 7: Losing a voucher. Enrolling in Housing First was estimated to lower the three-year mortality of homeless veterans with mental illness by 4.6 percentage points (Costantini, 2025). In the model, with the drug price and π_V held, I compare households with SMI whom Treatment First's revocation puts on the street with the same households had they kept their voucher: losing it raises three-year mortality by about 3.6 percentage points, the same order of magnitude. This matters for the experiments because the households whose voucher Treatment First revokes bear its most direct cost, and the model's estimate of what losing a voucher does to their survival is of the size Costantini measured.

[32] The rule sets the tenant's contribution at thirty percent of its income, but the voucher pays only up to the fair market rent, so a holder that rents a dwelling above it pays the difference itself, which raises its share above thirty percent, as it does in HUD's data.

[33] Bassuk et al. (1997) compare 220 homeless mothers with 216 low-income housed mothers, so both groups are poor and the odds ratios, 1.8 for a homeless female caretaker and 2.4 for a male one, separate the transmission of homelessness from the poverty around it; a ratio against the general population would credit the housing penalty with everything else that differs between poor children and the rest.

Table 4: Untargeted moments

Moment	Data	Model	Source
Block 1: Homelessness			
Share of homeless by age, 25–34 to 65+[a]	26.0, 27.2, 21.0, 18.4, 7.5	26.4, 37.9, 24.2, 10.7, 0.8	HUD PiT 2024
Employed share of the homeless	46	21.4	Meyer et al. (2024)
Income of the homeless, over average	19.1	6.5	Meyer et al. (2024)
Block 2: Who uses and who has a SUD			
Share of users by age, 26–34 to 65+	45, 19, 14, 12, 10	67, 16, 10, 5, 2	NSDUH 2023
Mean age of users, years	39.6	34.6	NSDUH 2023
Unemployment given a SUD	16.2	14.5	NSDUH 2024
Drug use with SMI over without, ratio	2.44	2.64	NSDUH 2024
Drug deaths as a share of deaths, 25–44	20.0	14.5	CDC WONDER
Block 3: Housing and vouchers			
Renters paying over 30% of income	43.2	58.7	SCF 2022
Rent over income, voucher holders	31.2	39.0	HUD PSH 2023
Voucher holders living on wages	28.0	39.2	HUD PSH 2023
Block 4: Income and wealth distributions, shares of the total			
Income, bottom 1, 5, 10, 20%	0.01, 0.31, 0.92, 2.71	0.01, 0.19, 1.00, 3.40	SCF 2022
Net worth, bottom 1, 5, 10, 20, 50%	−0.16, −0.28, −0.29, −0.23, 2.2	0.00, 0.00, 0.00, 0.27, 6.3	SCF 2022
Block 5: Drug price elasticities			
Participation (extensive margin)	−0.3 to −0.9	−1.32	Saffer and Chaloupka (1999), Grossman and Chaloupka (1998)
Intensive margin (users)	−0.3 to −0.8	−0.63	Grossman and Chaloupka (1998), Olmstead et al. (2015)
Block 6: Transmission			
Child of a SUD parent acquires a SUD, odds ratio	1.44, 2.51	1.46	Mellentin et al. (2016), Yoon et al. (2013)
Homeless adult had a homeless parent, odds ratio	1.8 to 2.4	6.23	Bassuk et al. (1997)
Children living with a parent with a SUD, %	4.6	5.8	NSDUH 2024
Block 7: Losing a voucher			
Three-year mortality, households with SMI put on the street, pp[b]	4.6	3.6	Costantini (2025)
Block 8: Treatment under a condition			
Abstinence, contingent over unconditional housing, ratio[c]	1.26	1.40	Milby et al. (2005)

Notes: percent, except the elasticities and the odds ratios of the transmission block, the ratio in Block 8, the mean age (years) and the SMI ratio in Block 2, and percentage points in Block 7; the model is the Housing First benchmark, except in Blocks 7 and 8, which apply Treatment First to it (notes b and c). [a]The share aged 25 to 34 is the target of T_0 in Table 3, so of this row only the other four bands are untargeted. [b]Data: Costantini's instrumental-variables estimate of how much enrolling in the Housing First programme of the Department of Veterans Affairs lowers the three-year mortality of homeless veterans with mental illness. Model: the three-year mortality gap between the households with SMI whom Treatment First's revocation forces into homelessness and the same households had they kept their voucher, with the drug price and π_V held. [c]Data: consecutive weeks of abstinence in the arm housed on condition of abstinence over the arm housed without it, homeless adults with a cocaine disorder in day treatment. Model: the recovery hazard of voucher holders with a SUD over that of housed households with a SUD and no voucher, in the first year of Treatment First.

Block 8: Treatment under a condition. Milby et al. (2005) randomised abstinence-contingent housing against housing without conditioning, among homeless adults with a cocaine disorder in treatment: the contingent arm reached 8.32 consecutive weeks of abstinence against 6.60, a ratio of 1.26. The model's counterpart is the recovery hazard of voucher holders, whose voucher is conditional under Treatment First, against housed households without a voucher, both with a SUD: 1.40 times in the first year of the condition, the horizon of Milby's trial, close to its ratio. This matters for the experiments because treatment is how Treatment First reaches the households that already have a SUD, and the model's response over that horizon is of the size the trial measured.

6 Experiments and results

Every experiment replaces Housing First, the benchmark, with Treatment First, the rule of Section 4.9, and keeps the programme rationed and the tax system as it is in the benchmark. Two objects can respond to the reform: the drug price, by clearing the drug market, and the probability of drawing a voucher π_V, by holding net government revenue at its benchmark level $\bar{\mathcal{R}}$, so that what the screen saves is spent on vouchers paid to others. The experiments differ in which of the two respond and in whether the externalities on children are present. There are three experiments, and each answers its own question. The first two compare steady states (Section 6.2), each in its own half of Table 5, and the third considers the transition (Section 6.4).

The first experiment asks whether moral hazard alone justifies conditionality. In economics, moral hazard is defined as the effect insurance has on the incentive to avoid the loss it covers: when part of the cost of a loss falls on someone else, the insured does less to avoid it, which makes the loss more likely and raises what it costs whoever shares in it (Arrow, 1963; Pauly, 1968). Housing First insures a household against losing its home when drug use leads to a SUD, the SUD to unemployment and unemployment to homelessness. Since the assistance does not depend on use, part of what using costs falls on the government's budget, and through it on others who fund the programme. This is the case critics of Housing First normally make.

To test whether moral hazard justifies Treatment First, I remove the children from welfare and from the parent's preferences, and make their productivity independent of what the parent did. A household's drug use then harms no one but itself and the budget, so any gain comes from correcting moral hazard rather than from correcting the externalities on the children. I then re-solve the model under Housing First with these switches, keeping every calibrated parameter fixed, which gives the benchmark without externalities. I introduce Treatment First into it in two ways, both holding Housing First's drug price fixed. First, π_V is also held at its Housing First level, so that only the screen acts, which shows what deterring users does to the users themselves. Second, π_V clears the budget, so the savings are spent on vouchers for others; this run measures the gain from correcting moral hazard's cost to the budget that critics of Housing First have in mind. If Treatment First lowers welfare in these cases, moral hazard alone is no reason to make the assistance conditional. For completeness, I also show the case in which the drug price clears together with π_V.

The second experiment introduces Treatment First in the full calibrated model, with the externalities on children, in three steps. Each step lets one more object respond, so that each isolates one thing. The first two repeat the first experiment's two runs, so that each has a counterpart without the externalities. First, the drug price and π_V are both held at their Housing First levels. Only the screen then acts, as in the first experiment's first run, and the one difference between the two runs is whether drug use harms children, so comparing them shows what the externalities do to the verdict. Section 6.3 then isolates the externality itself within each run, as the part of the newborn's gain its parent does not count. Second, π_V clears the budget, as in the first experiment's second run, so that the screen's savings are spent on vouchers for others. This shows what that spending is worth once the children are counted, with its externality separated the same way. Third, the drug price clears the drug market as well. The drug price comes last because its response is a pure general-equilibrium price effect: the screen lowers the demand for the drug, and the cheaper drug draws users back. With both objects responding, as they would after a real reform, this step is also the main experiment: whether Treatment First raises welfare. Table 5 also shows, for both experiments, the drug price clearing with π_V held.

The third experiment is the transition from Housing First to Treatment First in the full model, with both the drug price and π_V clearing the drug market and the budget, respectively, every period (Section 6.4). The steady states are decades apart, and the households that have a SUD when the reform lands can be sure of keeping their assistance only by entering treatment and stopping use, while the disorder itself clears only as fast as recovery allows. The transition establishes how large the early costs of the reform are, who bears them, and whether children born at the reform gain.

After the main transition, I explore alternative transitions in which households with a SUD cannot stop using (Section 6.5): the screen with no exemption for treatment from the first day of the reform, the same screen announced five periods before it starts, and the same screen with treatment alone keeping the voucher for its first five periods, each at two detection rates. Section 6.1 defines the welfare criterion all three use.

The screen. The detection rate p_{det} is the chance, each year, that a voucher holder who uses drugs is caught. It sets how much using costs a holder, and so how strongly Treatment First deters use: as it rises, more users stop, and those who keep using lose their voucher sooner. What the rate captures is how often a household is tested, which no programme reports. I therefore set it so that the screen deters use as much as a drug test with a severe penalty does in the closest evidence, the US military's random testing (Section 2). Among service members who stayed after the military began testing at random and dismissing those who failed, past-year marijuana use fell by about 61 percent of what it would otherwise have been (Mehay and Pacula, 1999).[34]

The model's counterpart applies Treatment First to the benchmark's households, holding the distribution, the drug price and π_V at their Housing First levels, and follows each user subject to the screen, that is, each user who holds a voucher or is eligible and offered one. With $p_{\text{det}} = 20$ percent, 62 percent of them stop when

[34] That is the raw comparison among those who stayed, which leaves out the users the testing dismissed. Mehay and Pacula's own estimates of the policy's effect, which allow for who left, are 4 to 16 percentage points of the 35.7 percent who used in 1980, a fall of 11 to 45 percent. At half the calibrated rate, 41 percent of the users subject to the screen stop, inside that range (footnote 36).

the screen is introduced,[35] and none starts.[36] A household that does not use, and is in treatment if it has the disorder, is never flagged, since a positive screen is confirmed in a laboratory before it has consequences.

The release rate ψ_x is set to 1/3, so that the expected exclusion is three years. That is the same period for which the Housing Choice Voucher programme already bars a household evicted from federally assisted housing for drug-related criminal activity (Code of Federal Regulations, Title 24, Section 982.553). The test cost c_{test} is set to 0.031 percent of average earnings in the economy, the equivalent of $35 a test. The Texas Legislative Budget Board's fiscal note for Senate Bill 11 puts a test at $35 (Legislative Budget Board, 2013), the upper end of the estimates, and taking the upper end again loads the experiment against Treatment First.

6.1 The welfare criterion

Every welfare number in this paper is a consumption-equivalent variation (CEV). For a newborn it is computed as follows. Write V^b for expected lifetime value under the benchmark and V^c for expected lifetime value under a counterfactual policy. Both are taken at birth, before any of the newborn's draws are realised, including the household it is born into. Each is taken under its policy's own distribution of newborns, which the policy moves, since a child's starting productivity depends on what happened to its parent.[37] The CEV Δ is the proportional change in non-housing and non-drug consumption, applied at every date and in every state, that makes a newborn under the benchmark as well off as the same newborn under the counterfactual:

$$V^b\big((1+\Delta)\,c\big) \;=\; V^c(c). \tag{30}$$

A positive Δ therefore means the counterfactual is worth that much of a permanent consumption gain, and a negative one that it costs that much.

Whose welfare. Welfare here is measured for newborns, and it includes childhood in the parent's household and adult life after it. For a child born in period t it is

$$W_t \;=\; \underbrace{\sum_{j=1}^{J_K} \beta^{j-1}\,(\xi(j)-1)\,\frac{\int_{\mathcal{S}} U\big(X_c(j,s)\big)\,\mathrm{d}\Phi_{t+j-1}(j,s)}{\Phi_{t+j-1}(j,\mathcal{S})}}_{C_t} \;+\; \underbrace{\beta^{J_K}\,\frac{\int_{\mathcal{S}} V_{t+J_K}(1,s)\,\mathrm{d}\Phi_{t+J_K}(1,s)}{\Phi_{t+J_K}(1,\mathcal{S})}}_{A_t}, \tag{31}$$

where $V_t(j,s)$ and $\Phi_t(j,s)$ are the value and the distribution of Section 4.10 and Section 4.13 in period t, the subscript being the date and j the age, and $X_c(j,s)$ is the bundle of (11) that a child shares with a parent of age j in state s. The state in both integrals is the parent's: a child at home has no state of its own, its bundle is its parent's, and the state it enters adult life with is drawn at J_K by $\mathcal{N}$.

The first term, C_t, is childhood. A household joins the workforce at $j=1$ with its child just born, so the households of age j in period $t+j-1$ are the surviving parents of the children born in period t, each child in its j-th year at home. The child's flow utility from the bundle it shares with its parent is averaged over the parents alive that year, weighted by $\xi(j)-1$, the child's weight in the household's own equivalence scale, and discounted to the child's birth. Its parent counts the same flow at $\omega_c(\xi(j)-1)$ in (24), so only ω_c separates the parent's valuation from the criterion's: the parent internalises the share ω_c of its child's childhood and leaves the share $1-\omega_c$ external, as Section 4.10 defines it.[38]

[35] The military figure is for any past-year use of marijuana, a broader measure than the model's, whose users have a disorder of any drug, marijuana among them. Therefore, the two are compared as proportional responses.

[36] At half the rate ($p_{\text{det}} = 10$ percent), 41 percent of the users subject to the screen stop on impact, about two thirds of the military's response. The results hold: in the long run, with the drug price and π_V clearing, Treatment First cuts drug use by 14 and homelessness by 6 percent, against 21 and 11 percent, respectively, at the calibrated rate, and a newborn is better off by 1.65 percent of lifetime consumption, against 2.57.

[37] The ex-ante expected newborn welfare is the standard criterion in this literature (Conesa et al., 2009; Guvenen et al., 2023).

[38] A child whose parent dies before J_K stays in the average: for the rest of its childhood it is valued as the average child of its cohort rather than sent into adult life at the parent's death, as the law of motion (29) sends it, because an orphan has no consumption or dwelling of its own in the model. Sending it into adult life at the parent's death instead, with the probability $\chi(j)$ of (24), scores Treatment First lower: adult life carries the flow value of being alive ℵ and childhood does not, so that convention credits a child for losing its parent early, and Treatment First keeps more parents alive.

The last term, A_t, is adult life, which begins at J_K: it is the average value at $j = 1$ of the households that enter in period $t + J_K$, placed by $\mathcal{N}$ as in (29), discounted by the J_K years spent at home. The parent's own value, which Section 6.3 sets beside it, carries no such discount: the parent is at $j = 1$ in the year its child is born, so the parent's value and the child's childhood are dated alike. The value at $j = 1$ in A_t is the one the child's own problem maximises, so the child's own child enters there at the weight ω_c.

Counting the child's childhood at its own weight, $\xi(j) - 1$, rather than at its parent's $\omega_c(\xi(j) - 1)$ is what brings the externality on the child into welfare; Section 6.3 separates that externality from the rest of the gain. In a steady state the dates drop out.

Two horizons are used, and they are the same object measured at different dates: a newborn on the date the policy is implemented, who lives through the transition, and a newborn in the new steady state.

6.2 The long run

Table 5 sets the long run under Treatment First beside the benchmark, and, in the first five columns, Housing First and Treatment First without the externalities.

Treatment First without externalities. To capture the no-externality case, I treat the child as appearing only as an adult: its parent does not weigh it, its productivity does not depend on what its parent did, and its childhood, the years its parent is 26 to 50, is not counted in welfare. Every other parameter keeps its calibrated value: the economy is not refitted, so that the comparison changes nothing but these three switches.

With the drug price and π_V held at their levels under Housing First, Treatment First cuts drug use by 28 percent and the share with a SUD by 27 percent, against Housing First with the externalities switched off, while homelessness falls by 5 percent (first two columns of Table 5). And yet, no household is better off under Treatment First, with newborns losing 1.17 percent of lifetime consumption as adults. These losses capture the value of the option removed from a rational, forward-looking agent.

Rebalancing the budget does not change the ranking. When π_V instead rebalances the budget at Housing First's drug price (third column), the screen's savings buy more vouchers. π_V therefore rises from 6.75 to 6.81 percent, and 58 percent of households gain, but only by 0.05 percent of consumption on average. The gain is small and spread thin, while the loss is concentrated on the users the screen takes out: 96 percent of households with a SUD are worse off, and newborns still lose 1.09 percent of lifetime consumption as adults (since childhood is not counted here). Correcting moral hazard's cost to the budget is still not enough to outweigh the loss of that option to households that choose optimally.

When the drug price is also allowed to clear the drug market (fifth column), newborns still lose 0.74 percent as adults. The change from the third column is the general-equilibrium effect of the drug price and not a correction of moral hazard: the screen lowers demand, the price falls by 4 percent, the cheaper drug draws users back, and more of them are caught and lose their vouchers, which the budget then gives to others.

Treatment First with externalities. Having ruled out moral hazard as a reason for sobriety conditions, I now test, in the three steps set out at the start of Section 6, whether the externalities are a reason to impose them. Each step is a column of Table 5. Here I follow what each step does to drug use, to homelessness and to the children, and give its welfare gain; Section 6.3 then takes each gain apart into the externality and the rest.

With the drug price and π_V both held at their Housing First levels (seventh column of Table 5), only the screen acts, as in the second column: drug use falls by 43 percent, homelessness falls by 8 percent, and the share of children in a shelter falls from 1.03 to 0.95 percent. In this partial equilibrium, the parents themselves lose, as they do without the externalities: the screen takes an option from them and gives nothing back, and they lose 1.19 percent of their own lifetime consumption, close to the 1.17 percent of the second column. But in contrast to the model without the externalities, a newborn gains 2.82 percent, because the child gains more than its parent loses. The children of the users the screen deters get more of their parents' consumption and housing, and more of them start adult life without the earnings penalty, while the parent

counts only the share ω_c of either. In the aggregate, these *budget and earnings externalities*, which Housing First leaves in place, come with no trade-off against the *childhood homelessness externality*: by cutting use, the screen keeps more children out of a shelter than it sends to one by putting their parents out.

Letting π_V clear the budget at Housing First's drug price (eighth column) spends the revenue the screen frees on vouchers for households that do not use, or use but are not caught: π_V rises to 9.24 percent, drug use falls by 42 percent, homelessness falls by 20 percent, and the share of children in a shelter falls by 19 percent. Now the higher π_V makes the parents gain as well, by 2.42 percent of their own lifetime consumption, and a newborn gains 6.45 percent, because the higher π_V houses more children. Drug use falls by about as much as with both held. Childhood homelessness falls because the additional spending on vouchers lifts other low-income households of child-rearing age that do not use drugs out of homelessness.

Letting the drug price clear instead, with π_V held at its Housing First level (ninth column), lowers the drug price by 6 percent. The revenue the screen frees then goes unspent: drug use falls by 20 percent, homelessness falls by 4 percent, and a newborn gains 0.98 percent.

Finally, letting both the drug price clear the drug market and π_V clear the budget gives the main experiment (last column). Under this configuration, Treatment First cuts drug use by 21 percent, from 8.92 to 7.07 percent of households, and the share of households with a SUD by 19 percent, from 3.83 to 3.08 percent of households. It also raises the share of households with a SUD who are in treatment, from 24.6 to 29.0 percent, because a household with a SUD that is in treatment and does not use passes the screen.[39] As users who lose their vouchers leave the programme, it costs less at a given π_V. Tax revenue also rises, mainly because more children start adult life with higher productivity, and to a lesser extent because fewer adults are unemployed owing to their SUD. Thus, π_V rises from 6.75 to 7.80 percent, by 16 percent, making staying clean of drugs even more desirable. At the same time, taking users out of the drug market lowers the drug price by 5 percent, which draws some households back.

In full general equilibrium, Treatment First also lowers homelessness by 11 percent, from 0.62 to 0.55 percent of households, for three reasons. First, households that stop using spend less on drugs, and thus more of them can house themselves. Second, less SUD means less unemployment and so less homelessness. Third, the vouchers Treatment First revokes from households with a SUD go, through a higher π_V, to other low-income households. For households aged 26 to 50, the years a child lives with them, homelessness also falls by 11 percent, with the share of children in a shelter falling from 1.03 to 0.92 percent.

These are long-run outcomes, but the transition does not qualitatively reverse them. One concern is that homelessness and childhood homelessness could rise in the first years after the reform, because the households that already have a SUD face losing their vouchers unless they enter treatment and stay sober, while the number with a SUD falls only as fast as recovery allows. As I show, this concern does not materialise: few households lose their vouchers, and those that stop using spend less on drugs and can afford to house themselves, so neither rises even in the first year (Section 6.4).

A newborn is better off under Treatment First by 2.57 percent of lifetime consumption, and its parent by 0.69 percent of its own. The gain of 2.57 percent is below the 6.45 percent of the second step, where the drug price is held, because the cheaper drug draws users back, which lowers π_V, and their use harms their children. The share of children living with a parent who uses drugs falls from 16.35 to 13.24 percent, and the share living with one who has a SUD falls from 5.82 to 4.75 percent.

Of the households that enter adult life at 26, 99.9 percent are better off under Treatment First than they would be in the same state under Housing First (Table 5). This is because an entrant has never had the disorder and holds no voucher, so the screen costs it nothing unless it chooses to use, while a voucher is easier to draw and the drug is cheaper. Across all households, 74 percent are better off and 3 percent worse off, and the rest are essentially unaffected. Finally, even 24 percent of those that have a SUD are better off.

[39] Under Treatment First a household with a SUD keeps its voucher by entering treatment and stopping. I also ran the experiment in which households with a SUD cannot respond at all: any use and the disorder itself fail the screen, in treatment or not, so treatment cannot help a household keep its voucher, although the government still pays for it. The results continue to hold: Treatment First then cuts drug use by 29 percent and the share with a SUD by 29 percent, homelessness falls by about as much as in the main experiment, by 11 percent, and a newborn gains at least as much. The results therefore do not depend on households with a SUD entering treatment and stopping.

Table 5: The long run under each policy

	No externalities					With externalities				
		Treatment First					Treatment First			
	Housing First	Both held	π_V clears	Price clears	Both clear	Housing First	Both held	π_V clears	Price clears	Both clear
Adults										
Drug price, benchmark = 1	1.03	1.03	1.03	0.99	0.99	1.00	1.00	1.00	0.94	0.95
Voucher probability π_V	6.75	6.75	6.81	6.75	6.85	6.75	6.75	9.24	6.75	7.80
Drug use	8.97	6.48	6.48	7.47	7.46	8.92	5.08	5.19	7.10	7.07
Carrying a SUD	3.85	2.82	2.82	3.25	3.25	3.83	2.22	2.28	3.09	3.08
Homelessness	0.60	0.57	0.57	0.59	0.59	0.62	0.57	0.49	0.59	0.55
SUD share of the homeless	20.6	15.1	15.1	16.8	16.8	20.2	12.6	11.9	15.9	15.6
In treatment, of those with a SUD	24.7	29.0	29.0	29.0	29.0	24.6	29.3	28.7	29.0	29.0
Children										
Living with a parent who uses drugs	16.44	12.15	12.14	13.97	13.96	16.35	9.53	9.76	13.29	13.24
Living with a parent who has a SUD	5.85	4.37	4.37	5.01	5.00	5.82	3.43	3.56	4.76	4.75
In a shelter	1.01	0.96	0.95	0.98	0.98	1.03	0.95	0.83	0.98	0.92
Welfare										
CEV of a newborn, percent of lifetime consumption	–	−1.17	−1.09	−0.89	−0.74	–	2.82	6.45	0.98	2.57
Share of the newborn's gain that comes in childhood	–	–	–	–	–	–	86	59	102	64
Households entering adult life at 26 better off	–	0	1	0	18	–	0	100	5	100
All households better off	–	0	58	1	63	–	0	74	1	74
Households with a SUD better off	–	0	0	13	13	–	0	17	14	24

Notes: percent unless the row says otherwise; each column is a steady state. The first five columns are Housing First and Treatment First without the externalities, with the child appearing only as an adult. Under Housing First the drug price clears and π_V is the benchmark's. Under Treatment First, the drug price and π_V are both held at the first column's levels in the second column; the drug price is held and π_V clears the budget against the first column's net revenue in the third; the drug price clears with π_V held in the fourth; and both clear in the fifth. The CEV in the second to fifth columns is the newborn's value at 26, against the first column. Housing First, the sixth column, is the benchmark with the externalities. Treatment First screens with a detection rate of 20 percent. In the seventh column the drug price and π_V are both held at their Housing First levels, in the eighth the drug price is held and π_V clears, in the ninth π_V is held and the drug price clears, and in the last both clear; when π_V is held, the revenue the screen frees goes unspent, and when it clears, π_V holds net government revenue at its benchmark level. Children are those living with households aged 26 to 50, the years a child lives at home. The CEV is against the benchmark. A household is better off when its value under Treatment First exceeds its value in the same state under Housing First, with households weighted by the Housing First distribution.

6.3 Where the gain comes from

A newborn's gain under Treatment First is the change in (31), and it has four parts, two of which are externalities. Write ΔC for the change in the child's welfare in childhood, the first term of (31), and ΔA for the change in its welfare in adult life, the second.

Childhood its parent does not count. The parent values the child's childhood at the share ω_c; the share $1-\omega_c$ of ΔC is the childhood externality, the budget and childhood homelessness externalities together: the consumption and housing a child gets when its parent spends less on the drug and is less often unemployed.

Better productivity at entry. A child's adult life changes for two reasons, which separate exactly. Hold the policy fixed at Treatment First and change only who enters adult life: fewer parents had a SUD, so fewer children carry the earnings penalty. This part, ΔE, splits as childhood does: the parent counts the share ω_c of it through the charge $D(y,d)$ in (24), so the share $1-\omega_c$ is the earnings externality.

The newborn's own adult life under the policy. Hold who enters fixed at Housing First's entrants, with the earnings penalties they carry, and change only the policy: the same child, on the same starting productivity, drawn under Housing First, lives its adult life under Treatment First, with a voucher easier to draw when sober and a cheaper drug, but with the screen in place. This is the effect of the policy on any adult, including its general-equilibrium effects when π_V or the drug price clear. It is exactly the gain the newborn's parent has, because the parent is an entrant too: ΔP, the change in the parent's own lifetime value when the policy switches, holding the distribution of entrants at Housing First's. The child receives it J_K years into its life, so it enters the newborn's gain as $\beta^{J_K}\Delta P$. This part is not an externality: it would be the same for a child whose parent never used.

What its parent does count. The share ω_c of ΔC and of ΔE: the part of the child's gain that its parent weighs in its own choices, so it is not an externality.

ΔP is the change in the parent's value at $j = 1$ of (24) when the policy switches from Housing First to Treatment First, averaged over the households that Housing First's newborns are born into, so that the same households are valued under both policies. ΔA is built from the same value at $j = 1$, the entrant's, discounted by β^{J_K} for a newborn, so the split of adult life is exact. For a newborn, $\beta^{J_K}\Delta P$ is the change in that value at a given starting productivity drawn under Housing First, so $\Delta E = \Delta A - \beta^{J_K}\Delta P$ is what the newborns' better starting productivity under Treatment First adds. In symbols,

$$\Delta W \;=\; \underbrace{(1-\omega_c)\,\Delta C}_{\text{childhood externality}} \;+\; \underbrace{(1-\omega_c)\,\Delta E}_{\text{earnings externality}} \;+\; \underbrace{\beta^{J_K}\Delta P}_{\text{own adult life}} \;+\; \underbrace{\omega_c\,(\Delta C + \Delta E)}_{\text{counted by the parent}} \;. \tag{32}$$

Table 6 reads each part off the benchmark's welfare curve, one after the other, so that the four sum to the newborn's CEV. Its last row adds the parent's own gain ΔP as a percent of the parent's own lifetime consumption, read off the curve for adult life, undiscounted, since the parent is at $j = 1$ when its child is born. The gain from the newborn's own adult life is smaller in absolute value than its parent's gain, for two reasons: the newborn receives it J_K years after birth, discounted, and its lifetime consumption, the base of the percent, includes its childhood as well as its adult life.

Table 6: Where the newborn's gain comes from

	Both held	π_V clears	Both clear
The newborn's gain ΔW, percent of lifetime consumption	2.82	6.45	2.57
Childhood its parent does not count, $(1-\omega_c)\Delta C$	2.20	3.47	1.50
Better productivity at entry its parent does not count, $(1-\omega_c)\Delta E$	1.06	1.00	0.46
The newborn's own adult life under the policy, $\beta^{J_K}\Delta P$	−0.75	1.54	0.43
What its parent does count, $\omega_c(\Delta C + \Delta E)$	0.31	0.44	0.19
The externality, the first two parts, percent of the gain	116	69	76
The parent's own gain ΔP, percent of the parent's lifetime consumption	−1.19	2.42	0.69

Notes: Treatment First against Housing First in the long run; the columns are the seventh, eighth and last of Table 5. The first row is the CEV of (30). The four rows under it sum to it: each is read off the benchmark's welfare curve after the ones above it. Childhood is the change in the child's flow at its weight $\xi - 1$, and better productivity at entry the change in its adult value from who enters; each is split into the share ω_c its parent counts and the share $1-\omega_c$ it does not. Better productivity at entry holds the policy and changes only who enters adult life; the newborn's own adult life holds who enters and changes only the policy, and equals the parent's own gain discounted to the child's birth. The parent's own gain is the change in its value at $j = 1$ when the policy switches from Housing First to Treatment First, with the distribution of entrants held at Housing First's, for the households Housing First's newborns are born into, as a percent of the parent's own lifetime consumption, undiscounted. It is larger in absolute value than the newborn's own adult life because the newborn receives the same gain J_K years later, discounted, and because the newborn's lifetime consumption, the base of its percent, includes its childhood as well as its adult life.

In the main experiment, the last column of Tables 5 and 6, the newborn gains 2.57 percent of lifetime consumption. Of that, 1.50 percent is childhood and 0.46 percent better productivity at entry, both of which its parent does not count. In all, three quarters of it is an externality. The rest is the newborn's own adult life under the policy, 0.43 percent, and the 0.19 percent that its parent does count, of childhood and of productivity at entry. Of the externality, the childhood part is the larger in every column: what the child gains at home accrues year by year for J_K years, while the earnings penalty reaches it once, at entry. The parents gain as well (the bottom row of the last column), by 0.69 percent of their own lifetime consumption, because the higher π_V and the cheaper drug outweigh the option the screen removes: in general equilibrium both the child *and* the parent gain.

With the drug price and π_V held, the first column, the externality is more than all of it, because the newborn's own adult life is a loss there, of 0.75 percent of lifetime consumption: the screen takes an option from every adult and gives nothing back, and the parents' own lifetime consumption falls by 1.19 percent. Recycling the savings, the second column, turns the newborn's own adult life into a gain of 1.54 percent and the parents' loss into a gain of 2.42 percent, since the vouchers go to households that do not use or are not caught; the cheaper drug then takes part of it back, in the last column.

6.4 The transition

To see what the reform does on the way to the long run, I compute the transition from Housing First to Treatment First. The transition starts from the benchmark, with the reform landing as an unexpected permanent shock at the start of the first year, and I simulate it for seventy years, all the way to the Treatment First steady state. Figure 2 shows what the transition looks like along eight key dimensions.

Panels (a) and (d) show drug use and the share of parents who use, respectively. Drug use falls on impact: 81 percent of its long-run fall happens in the first year. In the first year, the share of parents using drugs falls from 16.35 to 13.57 percent, above its long-run level of 13.24 percent. It stays above its long-run level until year 12, because it counts the parents who have a SUD when the reform lands, whose number falls only slowly (panel (b)). From then on it moves around its long-run level. The screen affects the decisions to go to treatment and to use drugs, both directly and through expectations. A voucher holder that uses risks losing its voucher within a year, and a household that uses risks having a voucher offer revoked later, when it may need one but have a SUD, since a disorder takes years to clear.

Panel (b) shows the share with a SUD. This share falls gradually, because recovery is slow: half of its long-run fall is done by year 11, and nine tenths by year 27. The households that still have a SUD keep demand for the drug up, and the drug price stays above its long-run level for 33 years (panel (c)). The elevated price deters households without a SUD, at every age: in the first decade the number of drug users without a SUD falls by up to 23 percent, against 18 percent in the long run.

Panels (e) and (f) show homelessness and the share of children in a shelter, respectively. Neither rises at any point. Homelessness falls from 0.62 percent of households to 0.60 percent by year 10 and 0.56 percent by year 69, against 0.55 percent in the long run, and the share of children in a shelter falls from 1.03 to 1.00 percent by year 10 and 0.93 percent by year 69, against 0.92 percent in the long run. Both could have risen, since the households that have a SUD when the reform lands face losing their vouchers unless they enter treatment and stay sober. They do not, for three reasons. First, few households lose their vouchers: one with a SUD keeps its voucher by entering treatment and not using, and the screen catches a user with a probability of only 20 percent a year, so at most 0.13 percent of households are excluded from the programme in any year. Second, drug use falls from the first year (panel (a)), so households spend less on drugs and have more left over to house themselves, and fewer acquire a SUD and experience longer unemployment. Third, after the first three years, when it is as low as 6.55 percent, π_V stays above its Housing First level of 6.75 percent, so the vouchers the screen frees go to other low-income households.

Panel (g) shows the CEV of each cohort by its year of birth. Children born at the reform gain 1.45 percent of lifetime consumption, less than the 2.57 percent of those born in the long run, for two reasons. First, they grow up while the share of households with a SUD is still close to its Housing First level, since it falls only as fast as recovery allows: from 3.83 to 3.47 percent in the first ten years, against 3.08 percent in the long run. More of them therefore spend their childhood with a parent who has the disorder, and more enter adult life with the earnings penalty. Second, π_V is lower in the first years than in the long run: between 6.55 and 6.90 percent in the first decade, against 7.80 percent. Part of what Treatment First adds to the budget is the taxes paid by adults who grew up without the earnings penalty, while those adults start working only J_K years after the reform. Until then, their parents are less likely to draw a voucher. The gain rises towards its long-run level for children born later, whose parents were themselves born after the reform.

Of the adults alive at the reform, only 11 percent are worse off under it, and even of those who have a SUD when it lands, 33 percent benefit, because the lower drug price more than offsets the loss from the sobriety requirement for some of them. Panel (h) repeats the comparison in every period of the path, with households in each state weighted as under Housing First. The share worse off stays between 2 and 11 percent, and the share of those with a SUD who are worse off falls from 67 to 50 percent as the price comes down further. Of the households entering adult life at 26, 33 percent are better off in the first year, when π_V is still close to its Housing First level, and 99.9 percent by year 69.

6.5 Alternative transitions

The main transition lets a household with a SUD keep its voucher by entering treatment and not using. This section asks what the first periods of the reform look like when households with a SUD cannot stop using,